Deck Diagrams and Materials List – 31 ft x 21 ft Freestanding Deck

Blue Springs, Missouri | Companion to the July 15, 2026 construction plan

Prepared July 16, 2026

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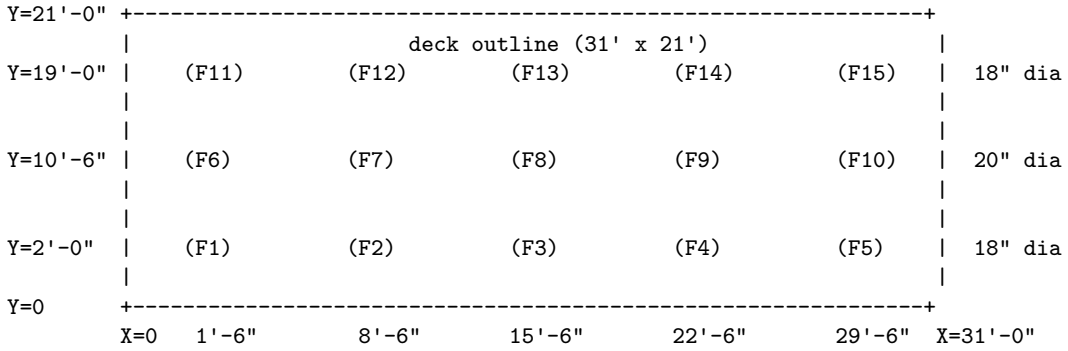
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All coordinates measure from the **finished deck corner (0,0)** – the southwest corner of the 31 ft x 21 ft decking surface. X runs along the 31-ft side, Y along the 21-ft side. The FRAME is inset 1 1/2 in on all four sides (30'-9" x 20'-9" outside of rims) so the 2x10 fascia finishes the deck at exactly 31 ft x 21 ft. Dimensions in the text govern over the sketches.

1. Pier and Footing Diagrams

1.1 Pier layout plan (15 piers)

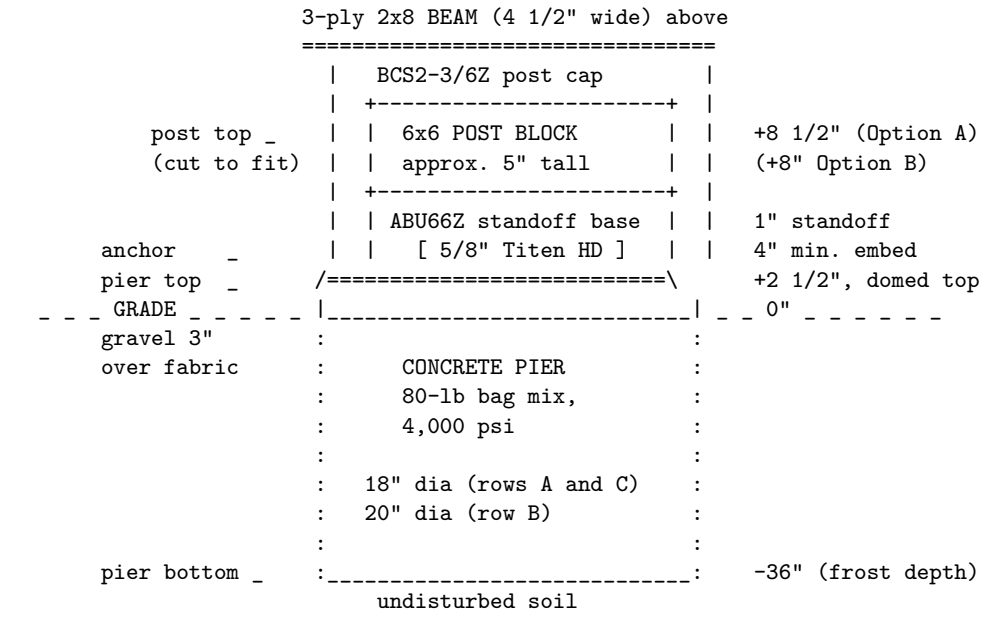


```
|--1'-6"--|---7'-0"---|---7'-0"---|---7'-0"---|---7'-0"--|--1'-6"--|
row spacing (Y): 2'-0" / 8'-6" / 8'-6" / 2'-0"
```

Row	Y	Piers	Diameter	Bottom	Top	Post load (max)
Beam A	2'-0"	F1-F5	18 in	36 in below grade	2 1/2 in above grade	2,100 lb
Beam B	10'-6"	F6-F10	20 in	36 in below grade	2 1/2 in above grade	2,975 lb
Beam C	19'-0"	F11-F15	18 in	36 in below grade	2 1/2 in above grade	2,100 lb

Diameters assume 1,500 psf soil – confirm bearing value, diameter, and the 36-in frost depth with Blue Springs (816-228-0118) before digging.

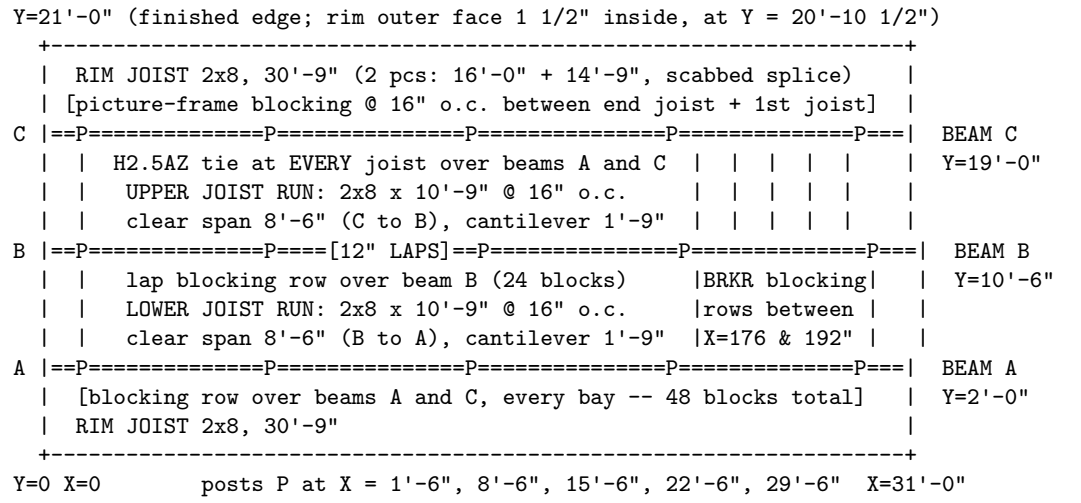
1.2 Single pier section (typical, all 15)



Per-pier concrete: 18-in pier = 5.3 cu ft (approx. 9 bags); 20-in pier = 7.0 cu ft (approx. 12 bags). Total 153 bags minimum; buy 168 (with 10%). Optional per pier: two #4 vertical rebar, 36 in long.

2. Joist and Beam Framing Diagrams

2.1 Framing plan



```
Joist lines (25): X = 2 1/4", 16", 32", ... 368", 369 3/4"
End joists at both 21-ft ends; beams run X = 2" to 30'-10" beneath them.
```

2.2 Beam build-up (each of 3 beams identical, 30'-8" long)

```
X=2"      8'-6" post        15'-6" post        22'-6" post        X=30'-10"
|=========================|=================================|  ply 1 (2x8x16
|        15'-4"             ^         15'-4"                   |  cut 15'-4")
|=============|============================|==================|  ply 2
|   8'-4"      ^         14'-0"             ^        8'-4"      |  (10'+16'+10')
|=========================|=================================|  ply 3
|        15'-4"             ^         15'-4"                   |
     ^ = ply splice, ALWAYS over a post; adjacent plies on different posts
Nailing: 10d x 3" HDG ring-shank, 2 rows @ 16" o.c. each face, staggered,
+ 3 nails within 6" of every ply end. Posts at 7'-0" centers; 1'-4" cantilevers.
```

2.3 Joist lap detail at beam B (section along a joist line)

```
       lower run joist                  upper run joist
  ========================]  <- 12" lap, (4) 10d face nails
                   [=========================
               [######################]
               [  3-ply 2x8 BEAM B    ]   toe-nail each joist (2) 10d
               [######################]   + 2x8 blocking in every bay
  each joist extends 6" past the beam centerline (Y=10'-6")
```

3. Decking Diagrams

3.1 Decking plan (picture frame + breaker)

```
Y=21' +-------------------------------------------------------------------+
      | BORDER 5/4x6 or 2x6 (runs full 31', parallel to field)   miter \ |
      |--+-----------------------------+---+-----------------------------+--|
      |B |  FIELD RUN 1: 43 courses    | C |  FIELD RUN 2: 43 courses    | B|
      |O |  boards 14'-9 9/16" long    | E |  boards 14'-9 9/16" long    | O|
      |R |  run parallel to 31' side   | N |  no butt joints in field    | R|
      |D |  gap 3/32" typ.             | T |  gap 3/32" typ.             | D|
      |E |  (9/32" at the two borders) | E |                             | E|
      |R |                             | R |                             | R|
      |  |  <-- boards this direction --|   |-- boards this direction ->|  |
      |--+-----------------------------+---+-----------------------------+--|
      | BORDER (runs full 31')                                  miter / |
Y=0   +-------------------------------------------------------------------+
      X=0    field starts X=5 19/32"    X=15'-6"                    X=31'
                                    CENTER BREAKER
                                    (5 1/2" wide, runs Y direction)
```

- 45 courses across 21 ft: 2 border + 43 field, all full 5 1/2-in boards – no rip.
- Border outside edge = finished 31' x 21' line, flush with the fascia face.
- Screws: 2 per bearing (16 in o.c.), 3/4-1 in from edges, ends predrilled. Option A: #9 x 2 1/2 in; Option B: #9/#10 x 3 in.

3.2 Edge section (dimension control)

```
deck surface +24" ->  [ border ][ field boards ...      1" (A) / 1 1/2" (B)
                      [========][====================
fascia face flush     |F|[ 2x8 RIM  ]  joist top +23" (A) / +22 1/2" (B)
with decking edge ->  |F|[  JOIST   ]
2x10 fascia,          |F|<- frame inset 1 1/2"; fascia restores exact 31'/21'
+13 3/4" to +23"      |F|   fascia: SS trim screws, 2 per 16"
```

3.3 Support under border and breaker

```
[end joist]  blocking @ 16" o.c.  [1st joist]      [joist @176"] blk [joist @192"]
|X=2 1/4"|===[12 1/4" long]======|X=16"|    ...    |==[14 1/2" long blocks]==|
   border + field-board ends                         field ends + BREAKER +
   both bear on this row                             field starts bear here
```

4. Materials List

All lumber pressure-treated SYP No. 2, ground-contact (UC4A). Quantities include the 10% waste/overage from the main plan (Section 10 there shows the calculations).

4.1 Structure (identical for both decking options)

Material	Use	Buy qty
2x8x12 GC	joists (50 pcs at 10'-9") + blocking	64
2x8x16 GC	beam plies (15) + rim joists (4)	21
2x8x10 GC	beam middle-ply end pieces	7
6x6x8 GC	15 post blocks approx. 5 in (A) / 4 1/2 in (B)	2
2x10x16 GC	fascia	7
2x10x8 GC	fascia end pieces	2
80-lb concrete mix (4,000 psi)	15 piers	168 bags
#4 rebar x 10 ft (OPTIONAL)	2 verticals per pier	10

4.2 Hardware and fasteners

Item	Model (example)	Qty
Standoff post base	Simpson ABU66Z	15
Anchor	Titen HD 5/8 x 6 in	17
Post cap	Simpson BCS2-3/6Z (often special order)	15
Uplift tie	Simpson H2.5AZ	55
Connector screws	Simpson SD9112R500 (500 ct)	2 boxes
Framing nails	10d x 3 in HDG ring-shank, 10-lb box	2
Rim nails	16d x 3 1/2 in HDG ring-shank, 5-lb box	1
Fascia screws	#8 x 2 1/2 in 305 SS trim-head, 5-lb box	1
End-cut preservative	Copper naphthenate 2%, quart	2
Beam tape (OPTIONAL)	G-Tape / Trex Protect 4 in x 65 ft	3 rolls

4.3 Ground preparation

Material	Qty
Woven landscape fabric, 4 ft x 100 ft	3 rolls
6-in fabric staples	1 box (100)
3/4-in clean crushed limestone, bulk	6.6 cu yd (approx. 9.3 tons)
Landscape edging (OPTIONAL)	104 lf

4.4 Decking – Option A (5/4x6)

Material	Qty
5/4x6x16 GC deck boards	103

Material	Qty
5/4x6x8 GC deck boards	4
#9 x 2 1/2 in exterior deck screws	approx. 2,700 (25-lb pail + 5-lb box)

4.5 Decking – Option B (2x6, recommended)

Material	Qty
2x6x16 GC	103
2x6x8 GC	4
#9/#10 x 3 in exterior deck screws	approx. 2,700 (25-lb pail + two 5-lb boxes)

Reminder: verify permit status, soil bearing, and frost depth with Blue Springs (816-228-0118) and call 811 before digging. Full calculations, cut lists, construction steps, and cost estimates are in the main plan (2026-07-15-blue-springs-deck-plan.pdf).